

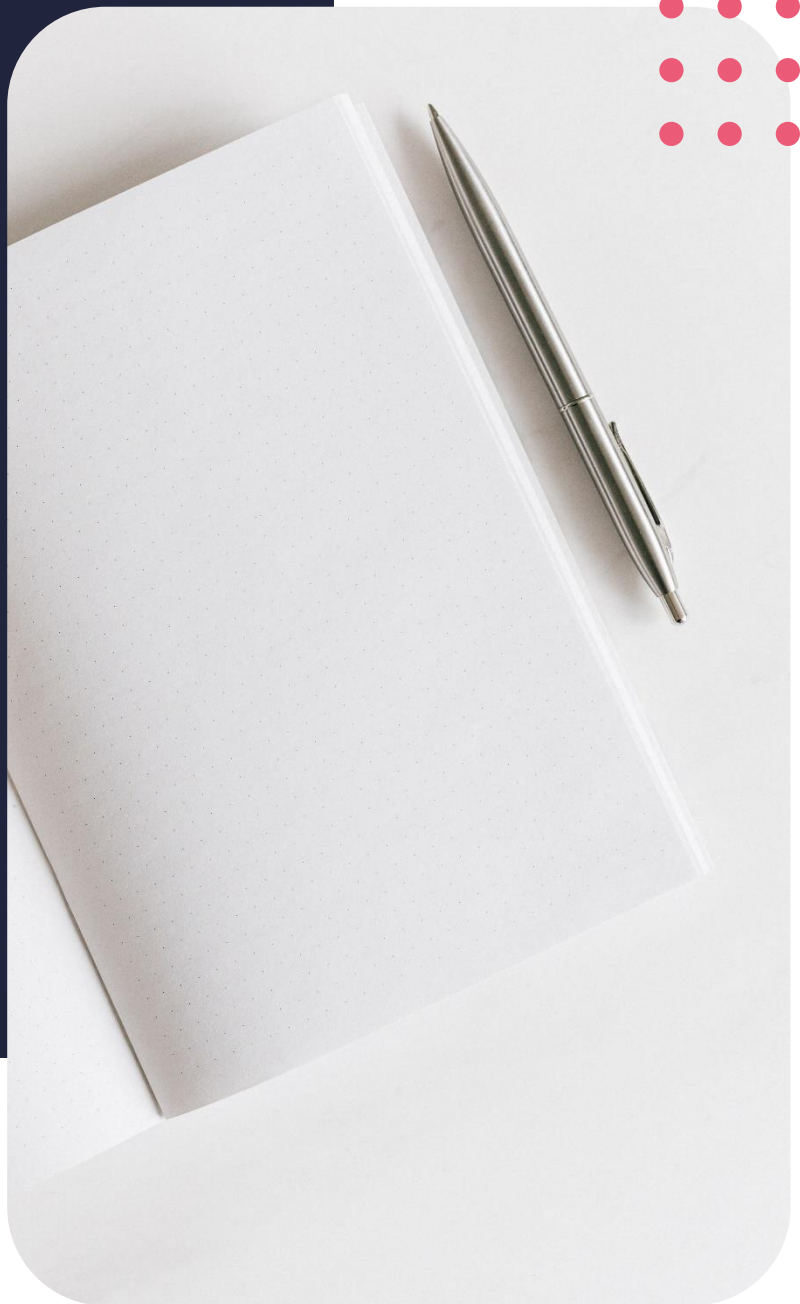
Diploma in Project
Management

Defining the Project Plan



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Lesson objectives

By the end of this lesson, you should be able to:

- Examine a project plan
- Introduce work breakdown structures (WBS)
- Explore project dependencies
- Examine estimation techniques
- Look at diagrams and resource planning

Introduction

Welcome to our third lesson of Project Management in which we unpack the second step in the life cycle – the project plan. The project plan is essentially a series of documents that define the execution and control stages of a project.

We have a jam-packed lesson today. Our first stop is to look at exactly what a project plan is. Then we will talk extensively about work breakdown structures. The third section of the lesson will focus on project dependencies and sequencing activities and then we will turn our attention to estimation techniques. The lesson will end with a look at various network diagrams and resource planning.

What is a project plan?

A project plan is a document used to guide & control the execution of the project. It clearly defines the goals of the project – the result that you wish to achieve. It outlines the plan to achieve the goals - the different steps that will need to be accomplished in order to achieve this goal. It may evolve as the project progresses – a project plan is more of a living document, as your project moves along, so your project plan will adapt to any changes that are made.

The project plan usually includes elements such as:

- A task list – a break down of work into little pieces so you can identify the scope and the time required for the project
 - Work breakdown structures visually show how the tasks are broken down into smaller activities
 - Dependencies outline – how the various tasks are linked and which ones can only start when a previous activity is finished.
 - Resource plan – this generally refers to staff members and their availability and how much time you will need them for
 - Project schedule – outlines the order in which things will be executed and how long the entire project will take.
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Work breakdown structures

We are going to spend a bit of time talking about work breakdown structures as these are key to the success of a project – basically a WBS defines all the things a project needs to accomplish organised into multiple levels and displayed graphically.

The first step to drawing up a work breakdown structure is to define the tasks that need to be done.

- Identify & document all activities that need to be performed in order to achieve the project goals
- This process promotes creating summary tasks which are a collection of & work packages (more about these three things just now...)
- Work packages are broken down into activities
- Activities are typically small segments of work that can be completed in a certain number of hours or days

WBSs are used to break projects down into more manageable sections. The sections are organised in a hierarchical structure. The further down a WBS you go more details become available - the biggest tasks or the most important things are at the top and then it breaks down into smaller subtasks. The further down a work breakdown structure you go, the more details are available. Deliverables are broken down into sub-deliverables & eventually work packages (which are the smallest amount of work that can be accomplished within a project)

Summary tasks

Summary tasks (or parent tasks) are high-level tasks. They assist with assigning task groups and determining the sequence in which things need to happen for your project to be successful. Basically, they are a presentation of the project milestones and/or phases in an easy-to-read format.

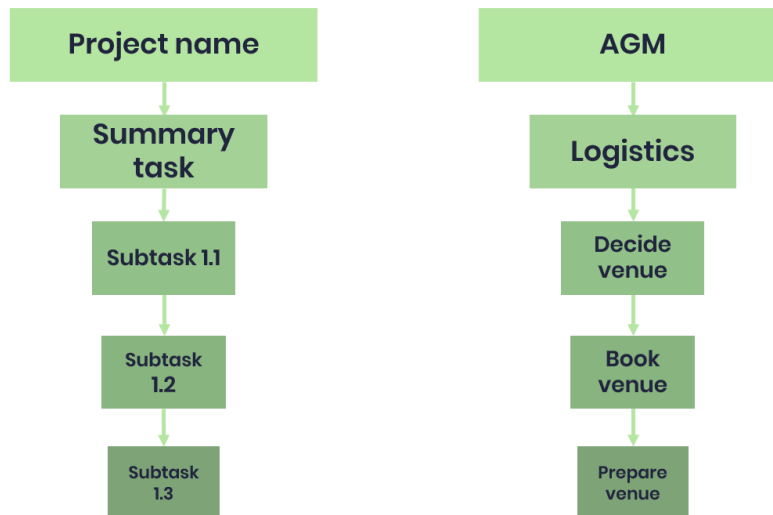
As you can see in this example, if the project is to hold an Annual General Meeting, three of the summary tasks would be to identify and communicate with all stakeholders, set the agenda and arrange the logistics such as the venue.



Sub tasks

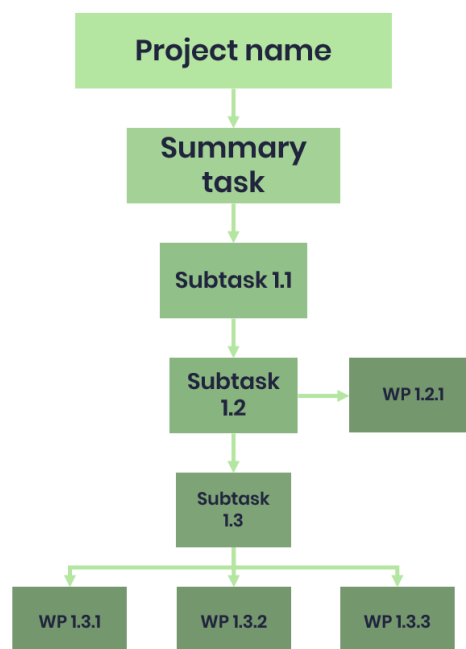
Sub tasks (or child tasks) are a more broken-down description of summary tasks – how the summary task is split up into different parts or activities.

Note the structure of subtasks all relating to the VENUE under the example summary task LOGISTICS for the Annual General Meeting project.



Work packages

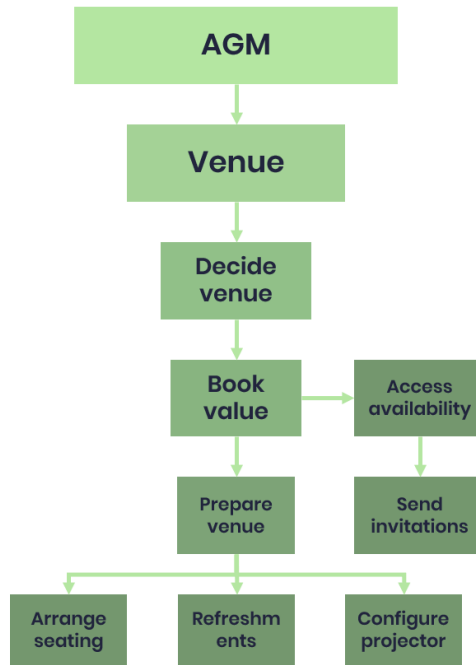
Work packages (or WPs) allow for large pieces of work to be broken down into smaller more manageable chunks. WPs enable multiple teams to focus on different aspects of the project that inevitably come together. When reviewing a WBS the work packages are the elements that cannot be broken down further. They are often grouped into areas of expertise (technology, human resources, finance) - one resource can be working on a work package under one summary task and later in the day they can work on another aspect that falls within their specialty group.



Work package rules

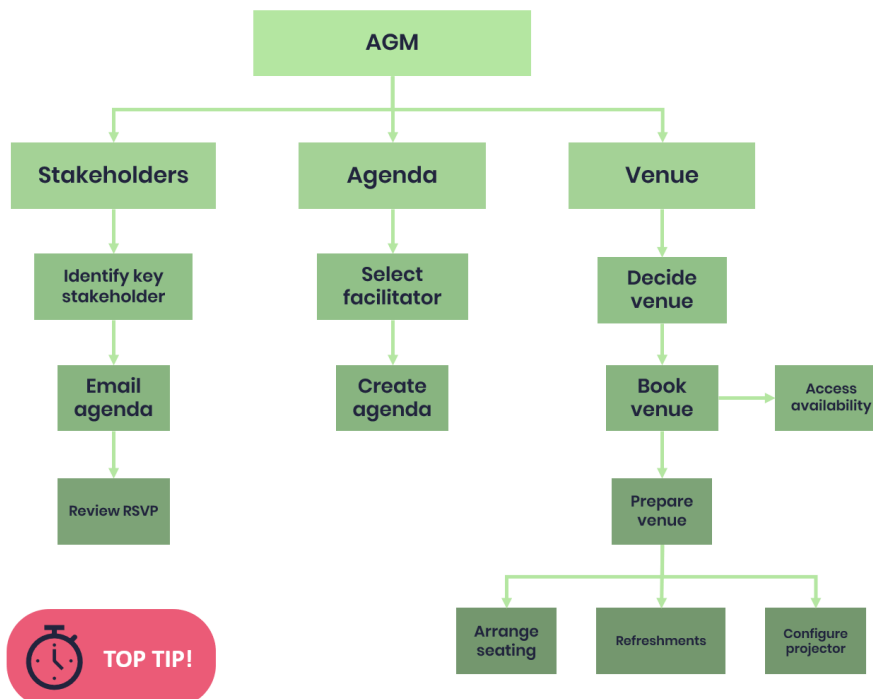
A few important rules for WPs before we look at the entire Work breakdown schedule diagram.

- They **should** not exceed 10 days of duration
- They **should** be independent of other work packages in the WBS
- Each work package is **unique** & should not be duplicated across the WBS



Top Tip

Here you have it – a Work Breakdown Schedule for arranging an Annual General Meeting consisting of Summary Tasks, Subtasks and Work packages. This is a useful sample that could be used as a basis for other projects.



Project dependencies

Dependencies in project management deal with managing and scheduling project tasks while keeping their sequences and requirements in mind. If task B requires the completion of task A, then we'll say that task B is dependent on task A. This may sound simple right now but in complex projects with several interdependent tasks, things can get messy.

Dependencies are the relationships of the preceding task to the succeeding task. Note that tasks may have multiple preceding tasks and multiple succeeding tasks.

Sequencing of activities

Once activities have been identified & documented, the next step is to identifying relationships between activities.

The four main **types of activity** dependencies include:

Finish-to-Start (FS) - Meaning the one activity must be completed before the succeeding activity can start

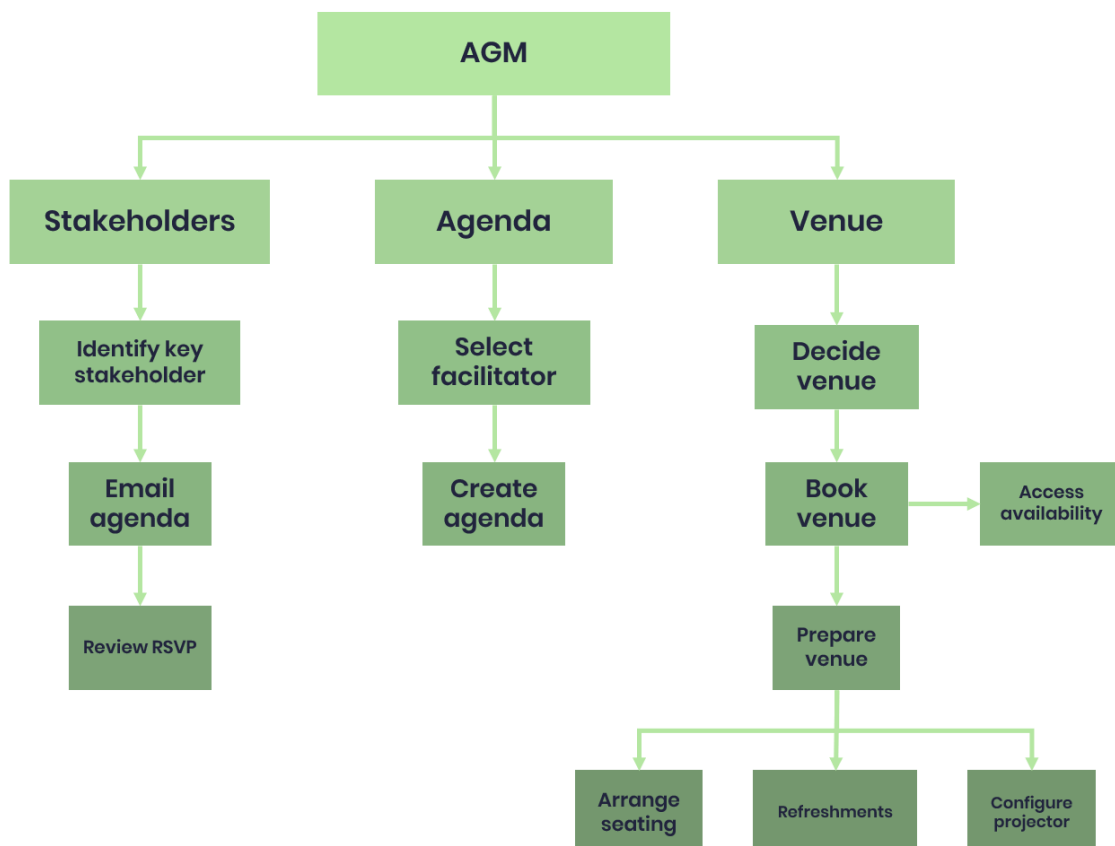
Start-to-Start (SS) - If two activities are connected through a start-to-start dependency, this means that the predecessor activity must have started before the successor activity can start.

Start-to-Finish (SF) - The start to finish relationship requires that a predecessor activity must have started before the successor activity can be finished. In practice, this type of dependency does not occur very often.

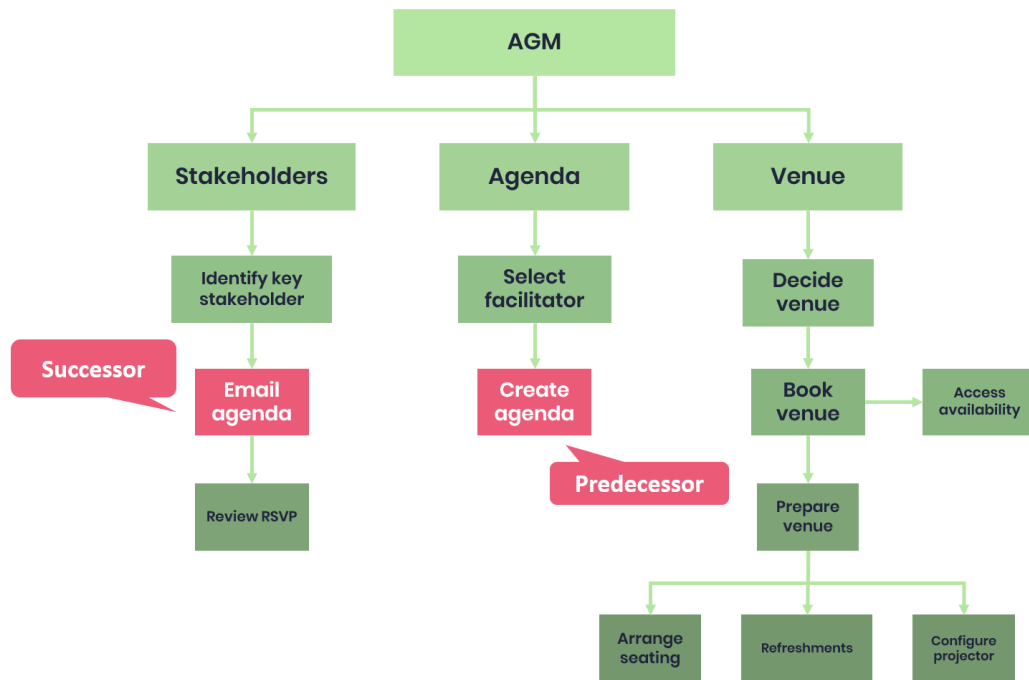
Finish-to-Finish (FF) - Under the finish to finish type of logical relationship, a successor activity requires the predecessor activity to be finished before it can be completed.

These relationships will then be applied to your Network Diagram.

So, if we look at the work breakdown schedule we drew up for our AGM again, perhaps now you can spot some dependency relationships between activities. Can you spot them?

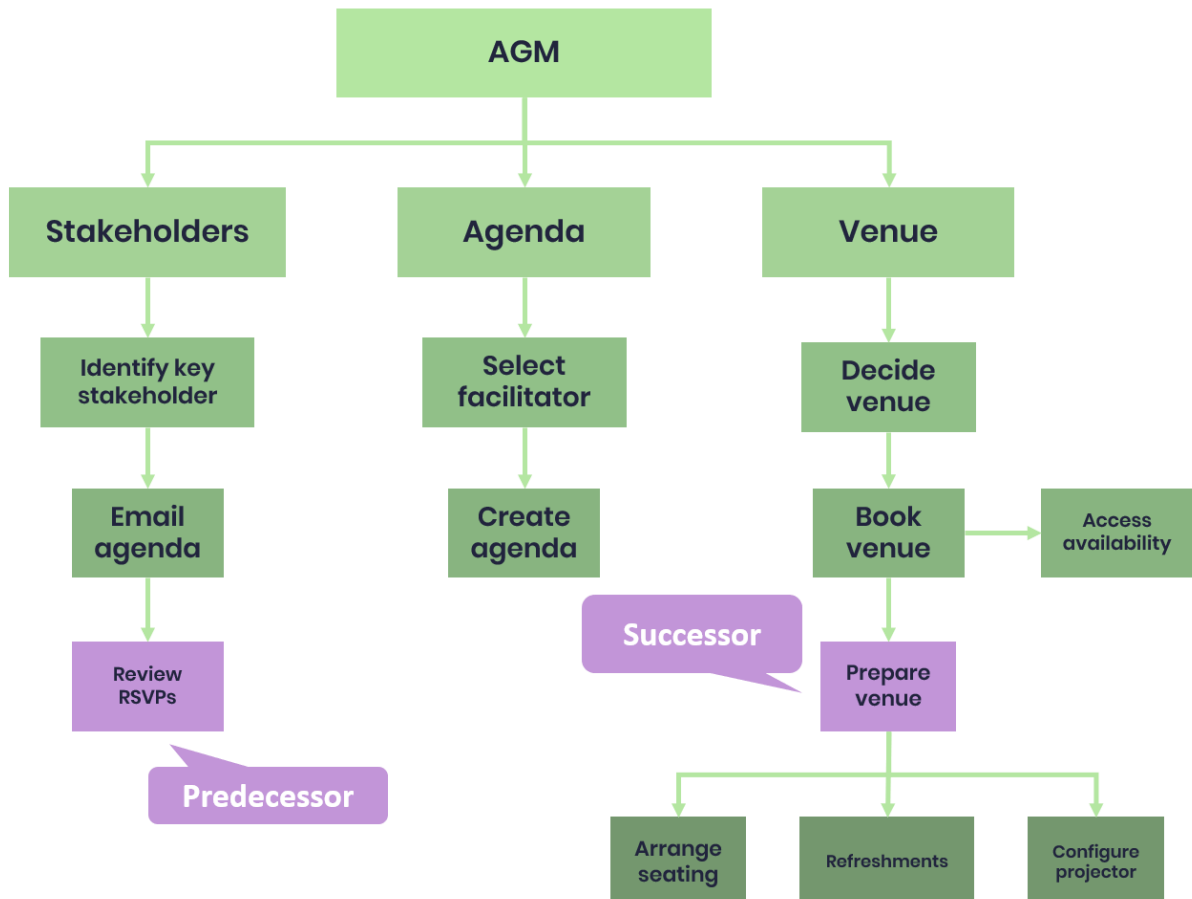


For example, the 'Create the agenda' task would clearly have to take place BEFORE the 'email agenda' task. These activities would therefore have a FINISH-START relationship.



And you would not need to know how many people are coming (have RSVP'd) in order to prepare the venue with the appropriate number of seats and refreshments.

Make sense? So, you can see how important dependencies are when it comes to managing and scheduling tasks.



Estimation techniques

Project estimation techniques help project managers accurately estimate essential elements, such as cost and scope, within their projects. We are going to look at three popular estimation techniques to measure activity durations:

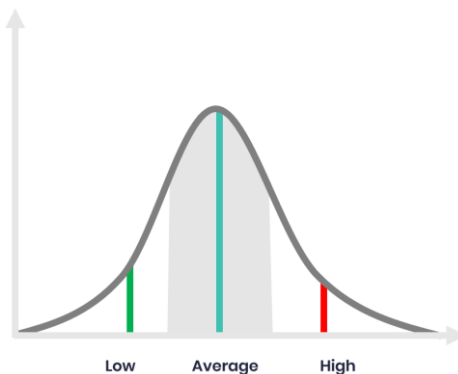
- Bell curve
- Delphi technique
- Bottom-up

Bell curve

According to Investopedia, A bell curve is a graph that is considered to be a normal distribution. The top of the curve shows the most likely event out of the data collected. After the mean is calculated, standard deviations are figured. Standard deviations that depict the returns of a security are known as volatility.

So, key takeaways here are:

- A bell curve is a graph depicting the normal distribution, which has a shape reminiscent of a bell.
- The top of the curve shows the mean, mode, and median of the data collected.
- Its standard deviation depicts the bell curve's relative width around the mean.
- Bell curves (normal distributions) are used commonly in statistics, and for analysing economic and financial data.



Delphi technique

The Delphi technique or method is a process used to arrive at a group opinion or decision by surveying a panel of experts. Several experts in the field are requested to complete questionnaires individually on the elements of the project. The answers are collated and then anonymously distributed to all the experts with an opportunity to revise their estimates. This process can occur many times before the final estimate will be based on the average estimate given by the participants.

Bottom-up technique

In a bottom-up approach to estimating, the project is broken down into smaller manageable sections to create a range of estimates based on the different tasks being performed. The staff resources that are responsible for the work will provide a time estimate. The overall estimate is calculated as an accumulation of the project's smaller tasks.

Helpful diagrams and resource planning

Our last stop today is to look at some practical planning tools to help you to manage your project effectively. We'll start with calendars and then focus on some useful diagrams. Then we will move onto resource plans and end with project schedules.

Project calendar

Project calendars are instrumental in planning the next day, week, month, quarter, or year. There are various ways to approach this but often two types of calendars are the most useful – a resource calendar and a company calendar.

Resource calendar

A resource calendar is used for planning, managing and monitoring resources, including both employees and equipment. It details working & non-working time of resources such as individual staff members. Non-working time includes: Vacation days, Public Holidays and Maternity/Paternity leave, for example

- Resources will vary depending on the project – can be anything from a meeting room, to a piece of machinery or a fleet of company cars.
- There are various very efficient resource scheduling software tools available including 10 000ft by Smartsheet and Float. As mentioned previously, our choice would be Monday.com.

Company calendar

Company calendars, like Google calendar, Microsoft and Teams enable individuals to share their commitments by detailing segments of time taken for meetings, tasks, conferences or workshops, leave etc.

When choosing a calendar make sure it is easy to use, simple to share, able to be customised to suit your needs and accessible on more than one platform.

Network diagram for setting up a website

A network diagram sets out the sequential and logical relationship between tasks. A diagram tool such as the example shown here is used to map out the schedule and visualise the work sequence for the project.

Activity	Description	Duration in hours	Team Member(s) assigned
A	Register domain name	2	Sue
B	Design website	10	Claire
C	Code website	36	Thabo
D	Web hosting	8	Rajesh
E	Copyright	3	Malika
F	Go live	3	Carlo

Task lists prioritising specific activities for certain individuals can be created from network diagrams, as can be seen here. Task lists are typically created at the start of a project and agreed upon by the entire team and stakeholders.

Did you know?

The absence of stakeholder buy-in can easily turn a multi-million dollar investment into a very expensive lesson about just how resistant to change we all can be. Stakeholder buy-in is the glue that binds all elements of a project together and ensures that the change will actually happen.

Develop a resource plan

To illustrate this we have used the example of creating a website again.

A good resource plan consists of a schedule that is as detailed as possible for the information known and the types of resources needed for each task.

Task	Name	Start Date	End Date	Duration
Register domain name	Sue	02/11/2020	02/11/2020	2
Design website	Claire	02/11/2020	03/11/2020	10
Code website	Thabo	04/11/2020	09/11/2020	36
Copyright	Rajesh	10/11/2020	10/11/2020	8
Host website	Malika	11/11/2020	11/11/2020	3
Go live	Carlo	11/11/2020	11/11/2020	3

As seen here, a good resource plan will have a single task owner for each task.

Developing a project schedule

Our final stop for today is to talk about project schedules. In short, project scheduling rounds up your plan, directs it, and gives you a comprehensive overview of progress. A project schedule can be used to:

- Monitor the status of project activities (using a review technique like RAG (remember that stands for RED, AMBER, GREEN))
- Implement changes/solutions for activities that have exceeded expected duration estimate
- Recognise potential risks & planning a preventative or corrective strategy

There are many software options available to help you to create smooth running schedules. A good project scheduling software consolidates all of the project info in a single collaborative space - it visualises project workflows, lets you prioritise tasks, manage resources, identify dependencies and bottlenecks (and solve them).

Some of the top scheduling software around include:

- Wrike which is great for teams of all sizes
- Hub Planner which is best for medium to large companies
- Celoxis is a really good all-in-one project scheduling platform
- TeamGantt is an ideal scheduling tool for beginners.

Conclusion

And that brings us to the end of our third lesson on project management. Today was all about the project plan. Your job as project manager is to ensure everything runs smoothly. We hope you found the tips, tools and techniques covered in this lesson interesting and useful.
